

NAME: SAMRENDRA NARAYAN SINGH

STUDENT ID: 47961171

UNIT: DATA AND VISUALIZATION FOR BUSINESS

SUBMITTED TO: DR. GENG SUN

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PART A

BUSINESS STORY

Cities of Southeast Asia and Australia are bustling with modern life and demands related to it, which can lead to individuals getting burdened and overwhelmed with various tasks. As a result the market and need for reasonable, reliable, and skilled assistance (help) is rising. Traditional options can often turn out to be unreasonably expensive, time-consuming, and lacking the personal touch that customers are looking for.

SOLUTION

Pro App aspires to revolutionize the connectivity between services and skilled tradespeople within their communities. It provides a reasonable and convenient for everyday tasks.

HOW Pro App WORKS

- I. CUSTOMER CREATES TASK
- II. TRADESPERSON (MAY/ MAY NOT HAVE CERTIFICATION) CREATE BID FOR TASK
- III. CUSTOMER EVALUATE BID
- IV. TRADESPERSON IS SELECTED
- V. TRADESPERSON ALLOCATED RELEVANT TASK
- VI. TRADESPERSON COMPLETES TASK
- VII. CUSTOMER PAYS TRADESPERSON ON COMPLETION OF TASK.

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IDENTIFYING THE ENTITIES

The entities that were identified from the business case include

- TRADESPEOPLE WITH ATTRIBUTES:
TRADESPERSON_ID INT PRIMARY KEY,
TRADESPERS_Name VARCHAR (100),
SKILL VARCHAR (100),
CERTIFICATION_STATUS BOOLEAN,
MEMBERSHIP_FEE DECIMAL (10, 2)
- CUSTOMERS WITH ATTRIBUTES:
CUSTOMER_ID INT PRIMARY KEY,
CUSTOMER_NAME VARCHAR (100),
CONTACT_INFO VARCHAR (255)
- TASKS WITH ATTRIBUTES:
TASK_ID INT PRIMARY KEY,
TASK_DESCRIPTION TEXT,
BUDGET DECIMAL (10, 2),
TASK_STATUS VARCHAR (50),
CUSTOMER_ID INT,

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FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMERS
(CUSTOMER_ID)

- TABLE BIDS WITH ATTRIBUTES:

BID_ID INT PRIMARY KEY,

TASK_ID INT,

TRADESPERSON_ID INT,

BID_AMOUNT DECIMAL (10, 2),

FOREIGN KEY (TRADESPERSON_ID) REFERENCES TRADESPEOPLE
(TRADESPERSON_ID)

- TABLE PAYMENTS WITH ATTRIBUTES:

PAYMENT_ID INT PRIMARY KEY,

TASK_ID INT,

PAYMENT_AMOUNT DECIMAL (10, 2),

PAYMENT_DATE DATE,

FOREIGN KEY (TASK_ID) REFERENCES TASKS (TASK_ID)

- TABLE CERTIFICATIONS WITH ATTRIBUTES:

CERTIFICATION_ID INT PRIMARY KEY,

TRADESPERSON_ID INT,

CERTIFICATION_TYPE VARCHAR (100),

CERTIFICATION_DATE DATE,

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FOREIGN KEY (TRADESPERSON_ID) REFERENCES TRADESPEOPLE
(TRADESPERSON_ID)

RELATIONS BETWEEN ENTITIES

Customer create Task (One to Many)

One customer can create many tasks.

Many tasks can be created by one customer.

The task has Payment (One to One)

One task has one payment upon completion.

Task have Bid (One to Many)

One task can have many bids.

Many bids can have one task.

Tradesperson make a bid (One to One)

One tradesperson makes one bid at a time.

Tradesperson have certification (One to Many)

One tradesperson can have many certifications

Many certifications have one tradesperson.

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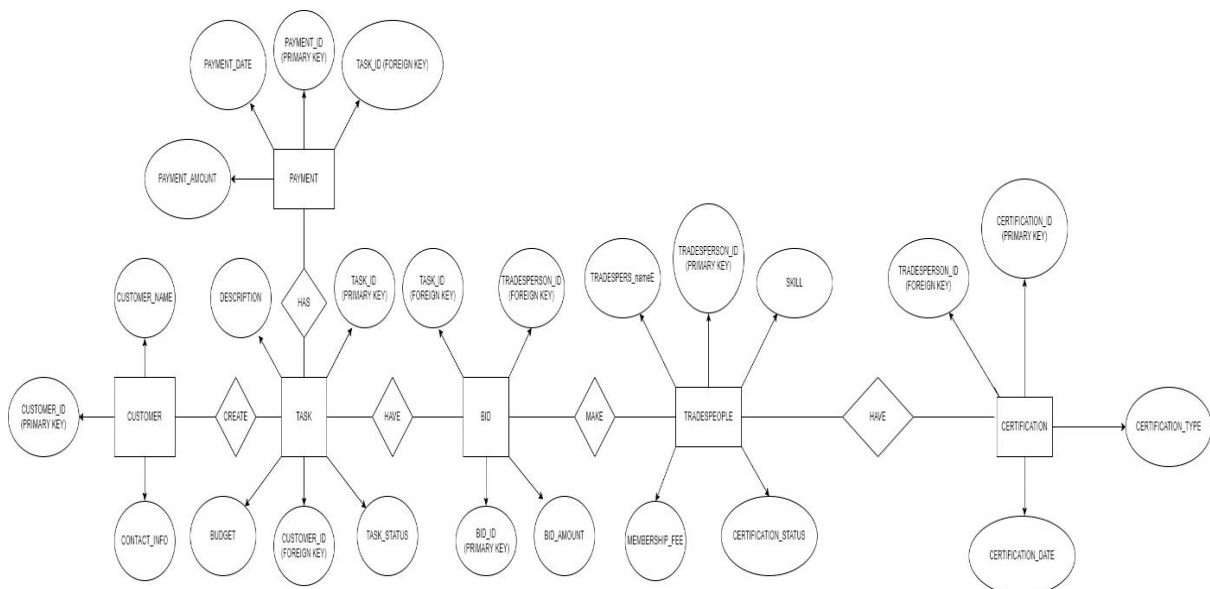
THE ENTITY RELATIONSHIP DIAGRAM (CHEN'S NOTATION)

Chen's notation represents entities, relationships, and attributes clearly and simply.

Rectangles represent entities.

Diamonds represent the relationships between entities.

Ovals represent the attributes of various entities.



MAKING THE DATABASE

```
Assignment 1
#Creating Database
1
2
3 • CREATE DATABASE ProApp;
4 • USE ProApp;
5
6 #Creating Tables
7
```

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CREATING THE TABLES IN THE DATABASE

```
Assignment 1 x
Limit to 1000 rows
8 • CREATE TABLE Tradespeople (
9     TRADESPERSON_ID INT PRIMARY KEY,
10    TRADESPERS_Name VARCHAR(100),
11    SKILL VARCHAR(100),
12    CERTIFICATION_STATUS BOOLEAN,
13    MEMBERSHIP_FEE DECIMAL(10, 2)
14 );
```

```
Assignment 1 x
Limit to 1000 rows
16 #Customers Table
17 • CREATE TABLE Customers (
18     CUSTOMER_ID INT PRIMARY KEY,
19     CUSTOMER_NAME VARCHAR (100),
20     CONTACT_INFO VARCHAR (255)
21 );
22
```

```
Assignment 1* x
Limit to 1000 rows
24 • CREATE TABLE Tasks (
25     TASK_ID INT PRIMARY KEY,
26     TASK_DESCRIPTION TEXT,
27     BUDGET DECIMAL (10, 2),
28     TASK_STATUS VARCHAR (50),
29     CUSTOMER_ID INT,
30     FOREIGN KEY(CUSTOMER_ID) REFERENCES customers(CUSTOMER_ID)
31 );
```

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```
Assignment 1 x
#Bids table
33
34 • CREATE TABLE BIDS (
35     BID_ID INT PRIMARY KEY,
36     TASK_ID INT,
37     TRADESPERSON_ID INT,
38     BID_AMOUNT DECIMAL(10, 2),
39     FOREIGN KEY(TRADESPERSON_ID) REFERENCES tradespeople(TRADESPERSON_ID)
40 );
```

```
Assignment 1 x
#Payments table
42
43 • CREATE TABLE PAYMENTS (
44     PAYMENT_ID INT PRIMARY KEY,
45     TASK_ID INT,
46     PAYMENT_AMOUNT DECIMAL(10, 2),
47     PAYMENT_DATE DATE,
48     FOREIGN KEY(TASK_ID) REFERENCES tasks(TASK_ID)
49 );
```

```
Assignment 1* x
#Certifications table
51
52 • CREATE TABLE CERTIFICATIONS (
53     CERTIFICATION_ID INT PRIMARY KEY,
54     TRADESPERSON_ID INT,
55     CERTIFICATION_TYPE VARCHAR(100),
56     CERTIFICATION_DATE DATE,
57     FOREIGN KEY(TRADESPERSON_ID) REFERENCES tradespeople(TRADESPERSON_ID)
58 );
```


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ENTERING DUMMY DATA INTO THE TABLES

```
Assignment 1 x
Limit to 1000 rows
60 • INSERT INTO tradespeople
61 VALUES
62 (1, 'John Doe', 'Electrician', TRUE, 100.00),
63 (2, 'Jane Smith', 'Plumber', FALSE, 75.00),
64 (3, 'Michael Johnson', 'Carpenter', FALSE, 120.00),
65 (4, 'Emily Davis', 'Painter', TRUE, 85.00),
66 (5, 'David Lee', 'Electrician', TRUE, 105.00),
67 (6, 'Olivia Taylor', 'Plumber', FALSE, 70.00),

Assignment 1 x
Limit to 1000 rows
68 (7, 'Benjamin Brown', 'Mason', FALSE, 130.00),
69 (8, 'Sophia Wilson', 'Roofer', FALSE, 90.00),
70 (9, 'Ethan Carter', 'Electrician', TRUE, 110.00),
71 (10, 'Ava Miller', 'Plumber', TRUE, 75.00);
72
73 #INSERT INTO customers
74 #VALUES
75 #(1, 'Alice Johnson', 'alicejohnson@gmail.com'),

Assignment 1* x
Limit to 1000 rows
73 • INSERT INTO customers
74 VALUES
75 (1, 'Alice Johnson', 'alicejohnson@gmail.com'),
76 (2, 'Bob Williams', 'bobwilliams@gmail.com'),
77 (3, 'Charlie Miller', 'charliemiller@gmail.com'),
78 (4, 'Diana Garcia', 'dianagarcia@gmail.com'),
79 (5, 'Ethan Moore', 'ethanmoore123@gmail.com'),
80 (6, 'Fiona Brown', 'fionabrown@gmail.com'),
```

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```
Assignment 1* x
Limit to 1000 rows
81 (7, 'Gabriel Hernandez', 'gabrielhernandez@gmail.com'),
82 (8, 'Isabella Clark', 'isabellaclark@gmail.com'),
83 (9, 'Chris Evans', 'chrisevans@gmail.com'),
84 (10, 'Jeffrey Archer', 'jeffreyarcher@gmail.com');
85
86 #INSERT INTO Tasks
87 #VALUES
88 #(1, 'Fix electrical wiring', 150.00, 'Open', 1),
```

```
Assignment 1* x
Limit to 1000 rows
86 • INSERT INTO Tasks
87 VALUES
88 (1, 'Fix electrical wiring', 150.00, 'Open', 1),
89 (2, 'Unclog drain', 80.00, 'Open', 2),
90 (3, 'Create new furniture', 3500.00, 'Open', 3),
91 (4, 'Paint exterior', 1200.00, 'In Progress', 4),
92 (5, 'Fix Refrigerator', 2500.00, 'Completed', 5),
93 (6, 'Repair house drainage', 1800.00, 'Open', 6),
```

```
Assignment 1* x
Limit to 1000 rows
94 (7, 'Fix house tiles', 4000.00, 'In Progress', 7),
95 (8, 'Repair roof', 150.00, 'Completed', 8),
96 (9, 'Upgrade electrical panel', 800.00, 'Open', 9),
97 (10, 'Install new water pump and purifier', 5000.00, 'In Progress', 10);
98
99 #INSERT INTO Bids
100 #VALUES
101 # (1, 1, 1, 140.00),
```

```
Assignment 1 x
Limit to 1000 rows
99 • INSERT INTO Bids
100 VALUES
101 (1, 1, 1, 140.00),
102 (2, 2, 2, 75.00),
103 (3, 3, 3, 200.00),
104 (4, 4, 4, 150.00),
105 (5, 5, 5, 135.00),
106 (6, 6, 6, 80.00),
```

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```
Assignment 1 x
Limit to 1000 rows
107      (7, 7, 7, 180.00),
108      (8, 8, 8, 160.00),
109      (9, 9, 9, 120.00),
110      (10, 10, 10, 90.00);
111
112      #INSERT INTO payments
113      #VALUES
114      #      (1, 1, 140.00, "2024-09-01"),
```

```
Assignment 1 x
Limit to 1000 rows
112 •    INSERT INTO payments
113      VALUES
114      (1, 1, 140.00, "2024-09-01"),
115      (2, 2, 125.00, "2024-09-05"),
116      (3, 3, 95.00, "2024-09-10"),
117      (4, 4, 175.00, "2024-09-15"),
118      (5, 5, 80.00, "2024-09-20"),
119      (6, 6, 110.00, "2024-09-25"),
```

```
Assignment 1 x
Limit to 1000 rows
120      (7, 7, 150.00, "2024-09-30"),
121      (8, 8, 75.00, "2024-10-01"),
122      (9, 9, 130.00, "2024-10-02"),
123      (10, 10, 120.00, "2024-10-03");
124
125      #INSERT INTO Certifications
126      #VALUES
127      #(1, 1, 'Electrician', "2023-01-15"),
```

```
Assignment 1* x
Limit to 1000 rows
125 •    INSERT INTO Certifications
126      VALUES
127      (1, 1, 'Electrician', "2023-01-15"),
128      (2, 2, 'Plumber', "2023-02-10"),
129      (3, 3, 'Carpenter', "2023-03-15"),
130      (4, 4, 'Painter', "2023-04-20"),
131      (5, 5, 'Electrician', "2023-05-25"),
132      (6, 6, 'Plumber', "2023-06-30"),
```

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Assignment 1*	
Limit to 1000 rows	
129	(3, 3, 'Carpenter', "2023-03-15"),
130	(4, 4, 'Painter', "2023-04-20"),
131	(5, 5, 'Electrician', "2023-05-25"),
132	(6, 6, 'Plumber', "2023-06-30"),
133	(7, 7, 'Mason', "2023-07-15"),
134	(8, 8, 'Roofer', "2023-08-20"),
135	(9, 9, 'Electrician', "2023-09-25"),
136	(10, 10, 'Plumber', "2023-10-05");

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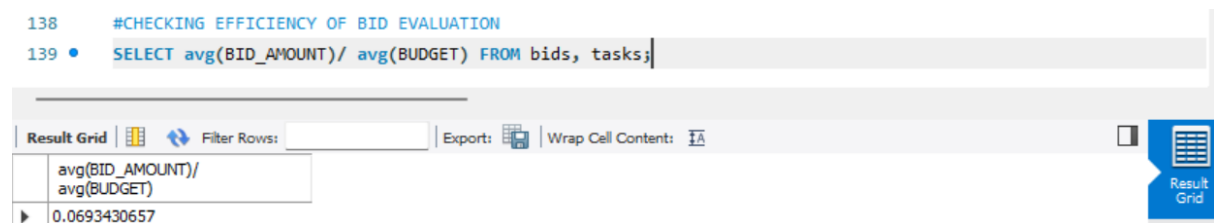
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PART B

ANSWERING IMPORTANT BUSINESS QUESTIONS AND PROVIDING RECOMMENDATIONS BASED ON THEM

Q1 HOW EFFICIENT IS THE EVALUATION OF BIDS?



The screenshot shows a SQL query editor with two lines of code. Line 138 is a comment: `#CHECKING EFFICIENCY OF BID EVALUATION`. Line 139 is a SQL query: `SELECT avg(BID_AMOUNT)/ avg(BUDGET) FROM bids, tasks;`. Below the editor is a 'Result Grid' toolbar with options for 'Filter Rows', 'Export', and 'Wrap Cell Content'. The result grid itself is a single-row table with two columns. The first column is labeled `avg(BID_AMOUNT)/ avg(BUDGET)` and the second column contains the value `0.0693430657`.

<code>avg(BID_AMOUNT)/ avg(BUDGET)</code>
0.0693430657

The above query calculates the ratio of the average bid amount to the average budget. This is useful to ascertain if the bids are overestimated or underestimated consistently.

The ratio in this case is a low value, which potentially means that bids are consistently getting underestimated.

This tells us that more work is required to find the right balance between the bid amount and the budget.

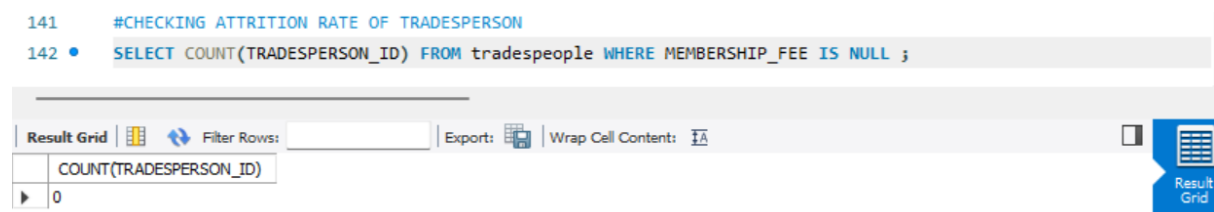
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Q2 WHAT IS THE ATTRITION RATE OF TRADESPEOPLE?



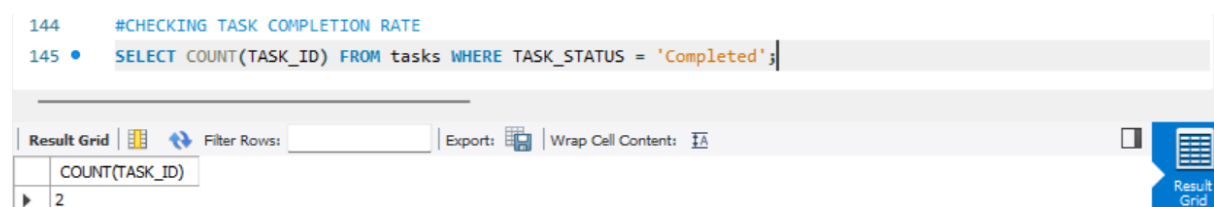
```
141 #CHECKING ATTRITION RATE OF TRADESPERSON
142 • SELECT COUNT(TRADESPERSON_ID) FROM tradespeople WHERE MEMBERSHIP_FEE IS NULL ;
```

COUNT(TRADESPERSON_ID)
0

The above query counts the number of tradespeople who have not paid their membership fee (attrition rate). In this case, the answer is zero, meaning each tradespeople has successfully paid their fee.

This tells us that the tradespeople are satisfied with the platform's proposition for them and have faced no significant challenges in the process.

Q3 HOW MUCH IS THE TASK COMPLETION RATE?



```
144 #CHECKING TASK COMPLETION RATE
145 • SELECT COUNT(TASK_ID) FROM tasks WHERE TASK_STATUS = 'Completed';
```

COUNT(TASK_ID)
2

The above query enumerates the number of completed tasks, which in this case is a low rate (2/10 or 20%) of task completion.

Hence work is required on issues like the assignment of tasks, quality of service, and communication among others.

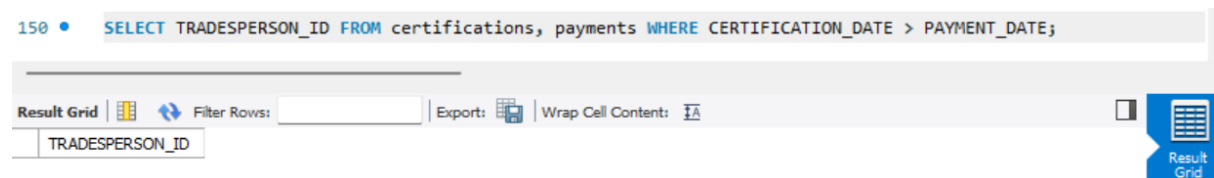
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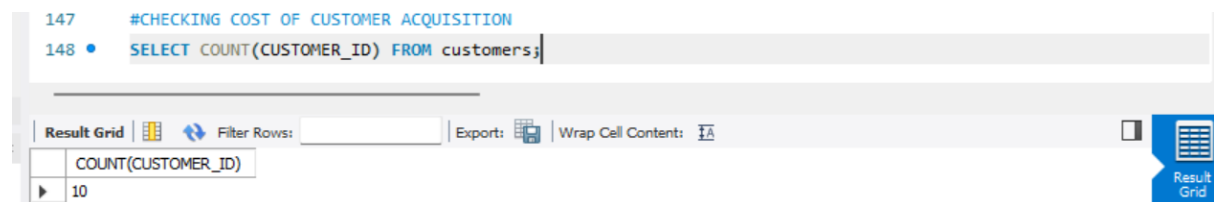
Q4 CHECKING FOR CERTIFICATION FRAUD



The above query checks for tradespersons who received certification after the payment date for tasks. The fact that no such occurrence was found means every tradesperson attended to the tasks after they received their certification.

This indicates that no certification fraud has occurred.

Q5 CHECKING THE COST OF CUSTOMER ACQUISITION



The above query counts the number of customer-id, which can be compared with marketing and similar expenses over time to understand the cost of customer acquisition.

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REFERENCES

INTERACTION WITH GEMINI AI FOR CREATING DUMMY (EXAMPLE) DATA